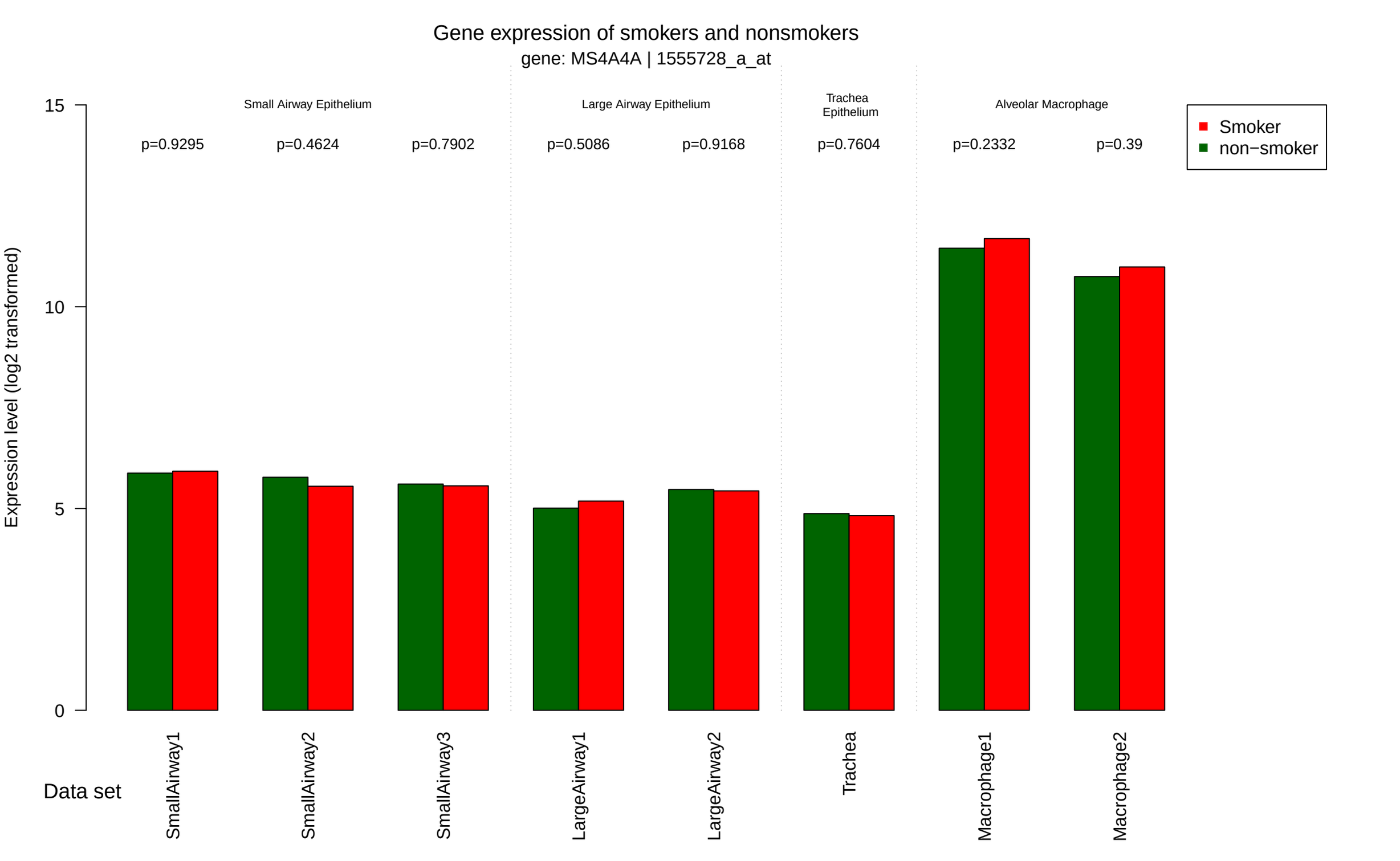
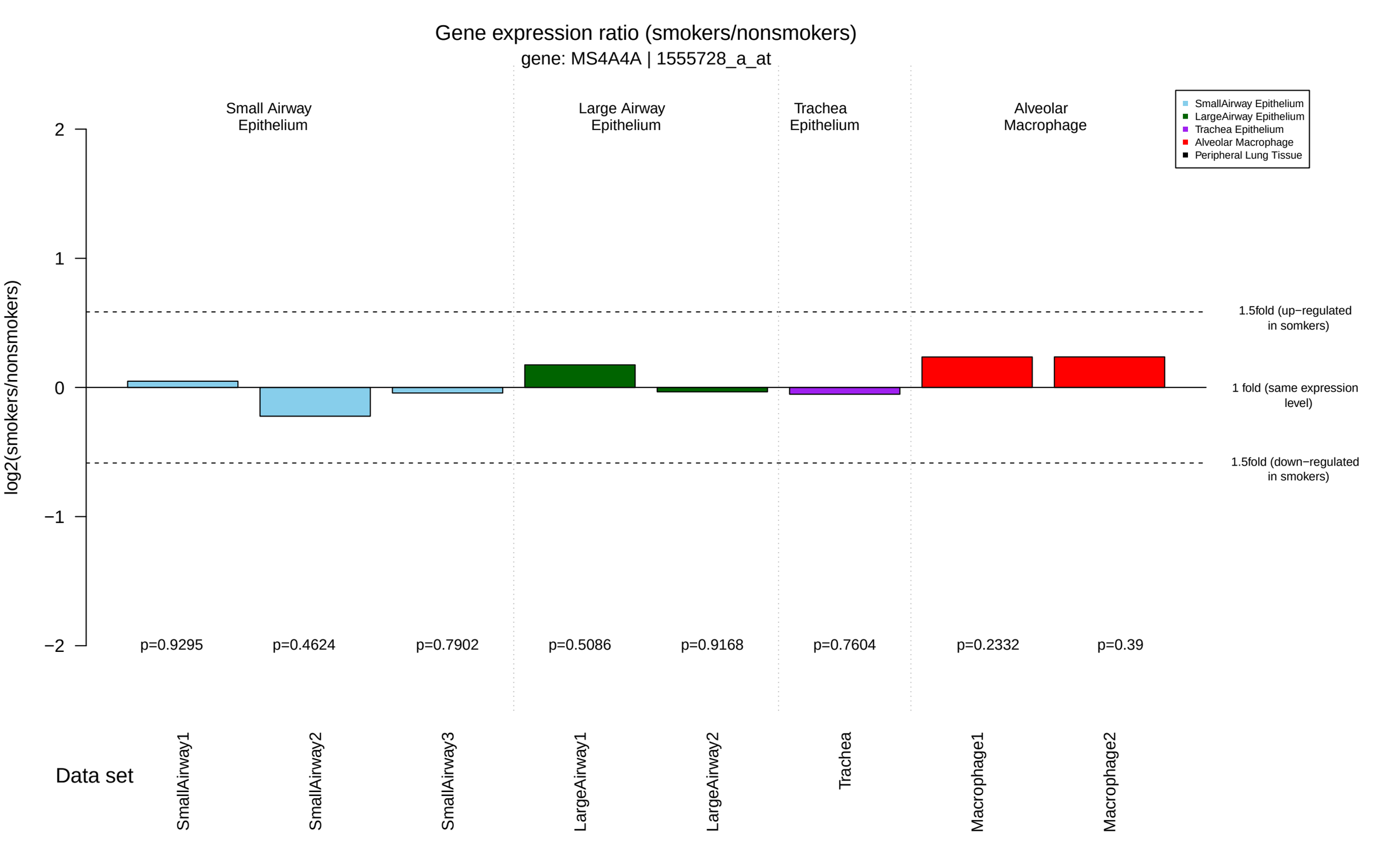
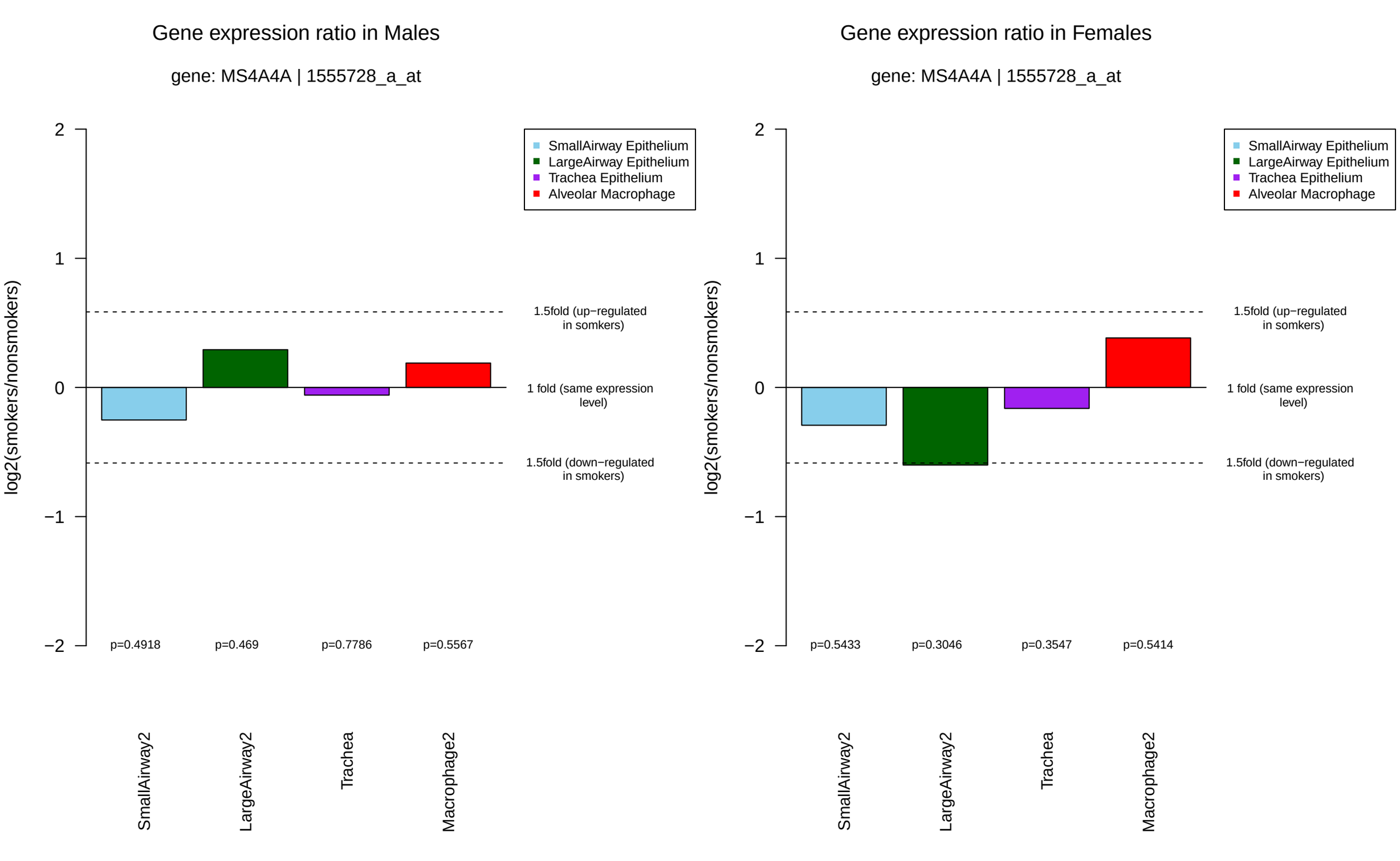
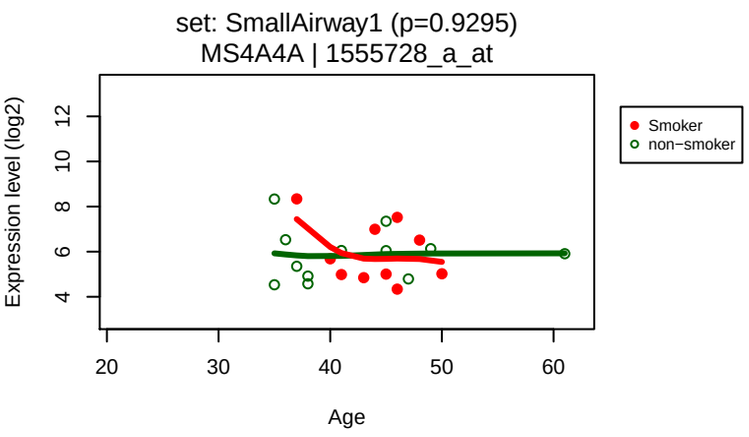
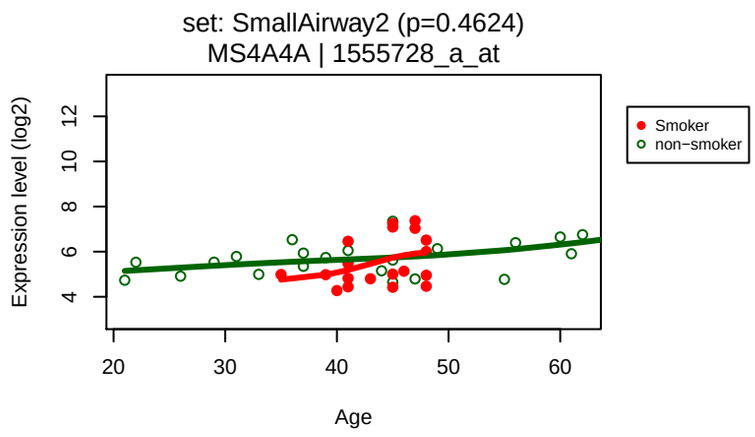
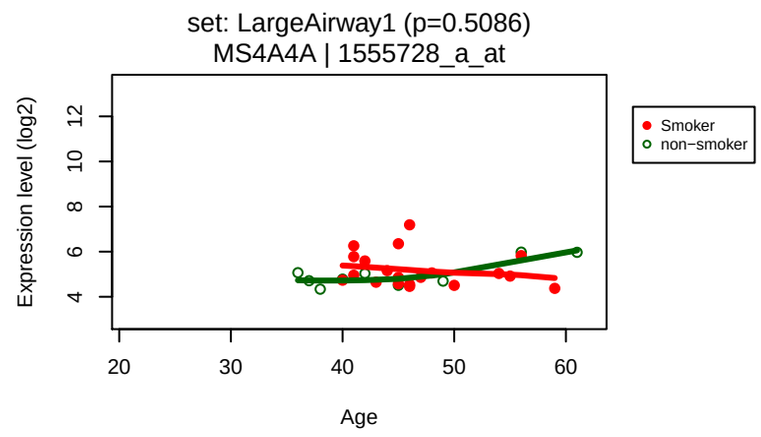

Gene expression of smokers and nonsmokers
gene: MS4A4A | 1555728_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9295
p=0.4624
p=0.7902
p=0.5086
p=0.9168
p=0.7604
p=0.2332
p=0.39
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

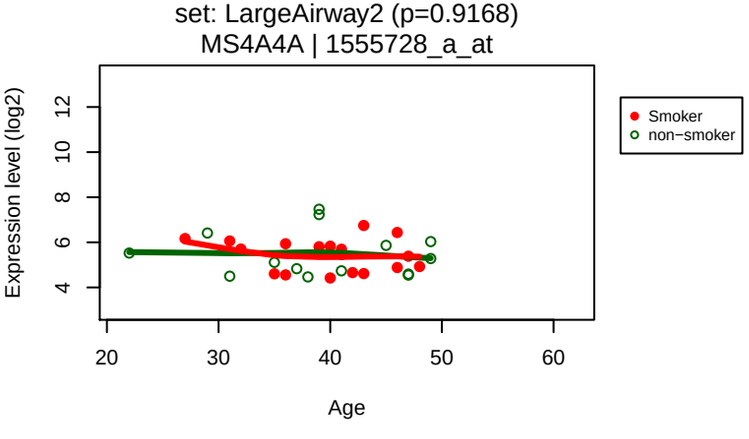

Gene expression ratio (smokers/nonsmokers)
gene: MS4A4A | 1555728_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9295
p=0.4624
p=0.7902
p=0.5086
p=0.9168
p=0.7604
p=0.2332
p=0.39
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

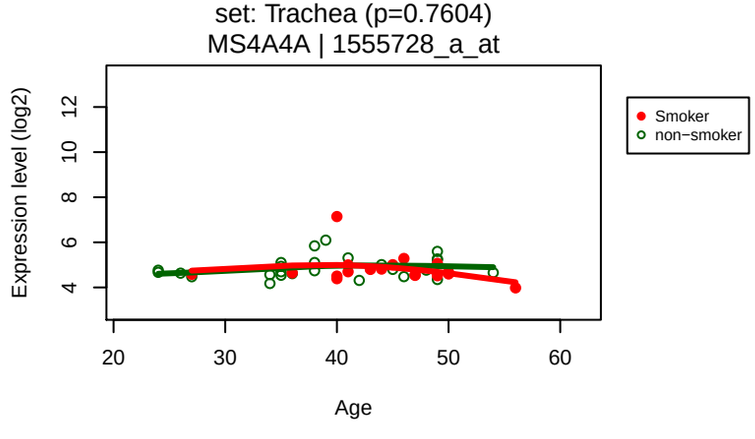

Gene expression ratio in Males
gene: MS4A4A | 1555728_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4918
p=0.469
p=0.7786
p=0.5567
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MS4A4A | 1555728_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5433
p=0.3046
p=0.3547
p=0.5414
SmallAirway2
LargeAirway2
Trachea
Macrophage2

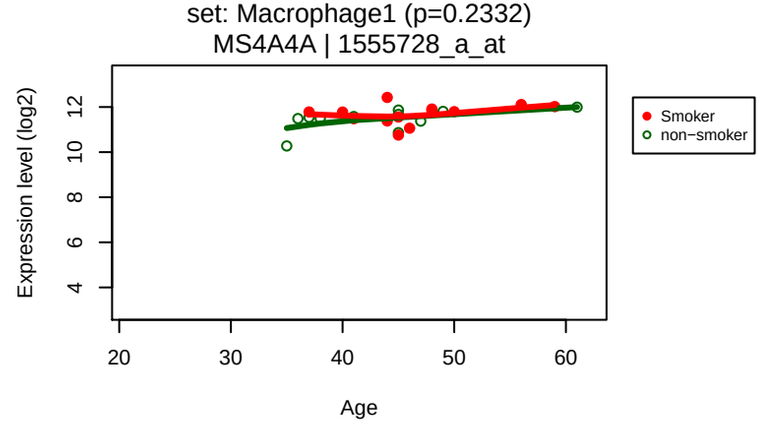

set: SmallAirway1 (p=0.9295)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker

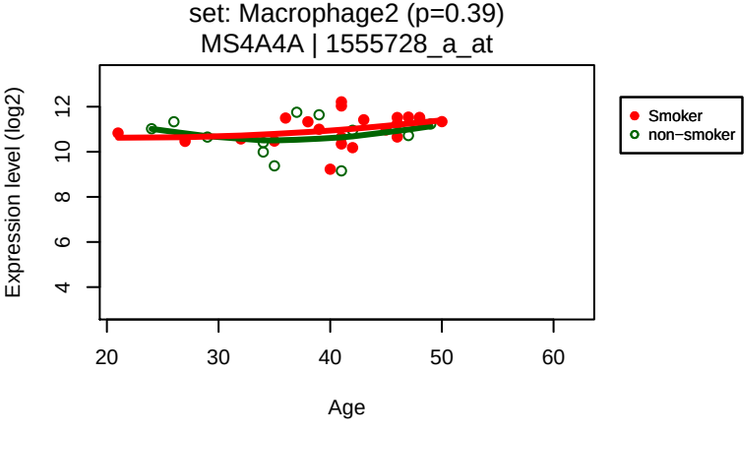

set: SmallAirway2 (p=0.4624)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker

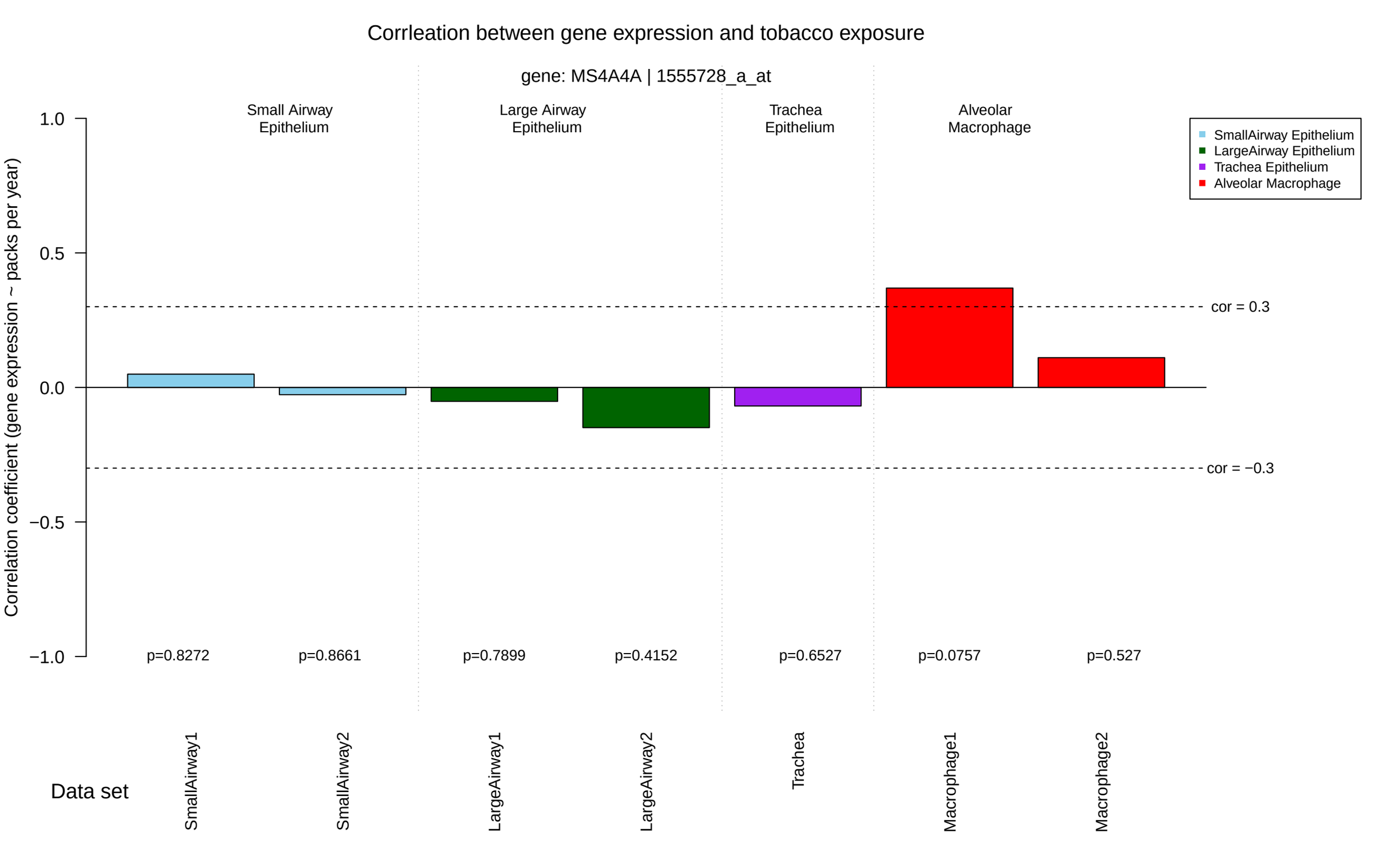

set: LargeAirway1 (p=0.5086)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9168)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7604)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2332)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.39)
MS4A4A | 1555728_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MS4A4A | 1555728_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.8272
p=0.8661
p=0.7899
p=0.4152
p=0.6527
p=0.0757
p=0.527
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2